

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION**CLASS IX • SOCIAL SCIENCE (SUBJECT CODE: 087) • ACADEMIC SESSION 2026-27****SET E — CASE STUDY & SOURCE-BASED SCENARIOS****CHAPTER 02: SHAPING OF THE EARTH'S SURFACE**

TIME ALLOWED: 45 MINUTES

MAXIMUM MARKS: 25 MARKS

SERIES: CBS-IX-SST-SET-E-02

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

General Instructions (SET E - Case Study & Source-Based Scenarios):

1. The question paper comprises **5 In-Depth Case Studies & Source Scenarios** with **5 MCQs each (Total: 25 Questions, 1 Mark each)**.
2. Read each source passage, historical inscription, geographic case, and economic scenario carefully before answering the corresponding questions.
3. Each question has 4 options with a single correct answer.
4. All questions are compulsory. There is no negative marking.
5. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

CASE STUDY 1: THE 2001 BHUJ EARTHQUAKE: TECTONIC RUPTURE & LIQUEFACTION IN KUTCH

On 26 January 2001, an intraplate earthquake of moment magnitude (M_w) 7.7 struck the Kutch district of Gujarat, with its epicentre located near Chobari village. The subterranean hypocentre (focus) was situated at a depth of approximately 16 kilometres along the South Wagad Fault. Seismological data revealed that compressional stresses accumulated as the Indian Plate continues to push northward into the Eurasian Plate at a rate of 4 to 5 centimetres per year. The resulting seismic rupture generated violent body waves (longitudinal P-waves and transverse S-waves), followed by destructive high-amplitude surface waves (Rayleigh and Love waves). In the Rann of Kutch and alluvial riverbeds, intense cyclic ground shaking triggered catastrophic soil liquefaction. Water-saturated, cohesionless sandy sediments lost their shear strength as pore-water pressure skyrocketed to equal overburden pressure, transforming solid ground into a boiling slurry. Thousands of sand boils and mud geysers erupted, causing multi-storey masonry structures to tilt and collapse abruptly.

Q.1 What is the fundamental distinction between an earthquake's focus (hypocentre) and its epicentre?

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| (A) The focus is on the Earth's surface, while the epicentre is in the clouds | (B) The focus is the underground point of initial fracture, while the epicentre is the point on the surface directly above the focus |
| (C) The focus measures water damage, while the epicentre measures wind speed | (D) The focus is only found in oceans, while the epicentre is only on land |

Q.2 What fundamental tectonic force is responsible for generating recurring seismic stress in northern and western India?

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|---|---|
| (A) Tensional pulling apart of the Indian plate from Africa | (B) Continuous convergent collision of the Indian Plate into the Eurasian Plate |
| (C) Subsidence of the Antarctic ice sheet into the ocean | (D) Gravitational pull of the Moon on subterranean magma chambers |

Q.3 What geotechnical phenomenon caused solid ground in the Rann of Kutch to behave like a fluid during the 2001 Bhuj earthquake?

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|---|--|
| (A) Chemical carbonation of limestone bedrock | (B) Soil liquefaction of water-saturated sandy sediments under cyclic seismic stress |
| (C) Thermal volcanic lava melting surface sands | (D) Glacial plucking of alluvial soil particles |

Q.4 Which seismic waves are recorded first by seismograph stations, and what is their wave propagation nature?

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|--|---|
| (A) S-waves; transverse shear waves that travel only through liquids | (B) P-waves; longitudinal compressional waves that travel through both solids and liquids |
| (C) Surface waves; circular water waves traveling along riverbeds | (D) Sound waves; sonic booms echoing through the stratosphere |

Q.5 Why were buildings on solid bedrock significantly less damaged during the Bhuj earthquake than buildings on unconsolidated alluvium?

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| (A) Bedrock absorbs seismic waves and dampens ground motion, whereas soft alluvium amplifies seismic wave amplitude | (B) Bedrock is softer than river sand and dissolves during earthquakes |
| (C) Buildings on bedrock were constructed with wooden straw roofs | (D) Seismic waves cannot enter solid rock layers at all |

CASE STUDY 2: FLUVIAL GEOMORPHOLOGY OF THE BRAHMAPUTRA RIVER & MAJULI ISLAND

The Brahmaputra River is one of the world's most dynamic fluvial systems, originating in the Chemayungdung glacier of Tibet (as the Yarlung Tsangpo) and cutting through the Eastern Himalayas via deep gorges before entering the Assam valley. In its upper reaches, steep hydraulic gradients produce dramatic V-shaped valleys, waterfalls, and rapid downcutting. However, upon debouching into the flat alluvial plains of Assam, the river's gradient flattens dramatically. Fed by torrential monsoon rains exceeding 2,500 mm and massive sediment loads resulting from young Himalayan landslides, the river transforms into an intensely braided channel network. In this braided plain lies Majuli, one of the world's largest inhabited riverine islands. Over the past century, excessive sediment deposition during summer floods has created transient mid-channel bars (chars), while centrifugal bank erosion along the outer cut-banks has reduced Majuli's land area from 1,255 square kilometres in 1901 to less than 400 square kilometres today.

Q.6 What geomorphic landform characterizes the upper youthful stage of the Brahmaputra in the Himalayas?

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|---|---|
| (A) Wide oxbow lakes and meandering floodplains | (B) Deep V-shaped valleys and narrow rocky gorges carved by vertical downward erosion |
| (C) Broad marine deltas with mangrove swamps | (D) Aeolian barchan sand dunes with slip faces |

Q.7 Why does the Brahmaputra develop a complex 'braided channel' pattern across the Assam valley rather than a single stable channel?

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|--|--|
| (A) The river has zero water discharge during the monsoon season | (B) A massive sediment load combined with an abrupt reduction in slope gradient forces the river to deposit mid-channel bars |
| (C) Local farmers construct concrete dams every five kilometers | (D) The river flows over frozen permafrost ground |

Q.8 How is Majuli island classified geomorphologically?

- (A) A volcanic island formed by oceanic hotspot eruptions (B) A large riverine alluvial island formed by fluvial deposition and multichannel braiding
- (C) A coral atoll created by marine polyps (D) A tectonic plate craton that drifted from Antarctica

Q.9 What hydraulic process causes the rapid shrinkage and bank collapse of Majuli island during monsoon floods?

- (A) Wind abrasion scouring granite boulders (B) Hydraulic bank scour and lateral erosion along the concave outer cut-banks
- (C) Chemical carbonation dissolving quartz crystals (D) Glacial plucking of frozen river sediments

Q.10 When a river meander on a mature floodplain becomes exaggerated and gets cut off during a flood, it forms:

- (A) A cirque (B) An oxbow lake
- (C) A yardang (D) A moraine

CASE STUDY 3: KARST GEOMORPHOLOGY IN THE BORRA CAVES OF THE EASTERN GHATS

Located in the Ananthagiri hills of the Araku Valley in Andhra Pradesh, the Borra Caves represent one of India's most spectacular karst landscapes. The caves are carved into Precambrian crystalline limestone (calcium carbonate, CaCO_3) by the perennial Gosthani River. Rainwater absorbs atmospheric carbon dioxide (CO_2) and soil organic acids, forming weak carbonic acid (H_2CO_3). As this acidic water percolates through vertical joints and bedding planes in the limestone, it chemically dissolves the rock via carbonation: $\text{CaCO}_3 + \text{H}_2\text{O} + \text{CO}_2 \rightarrow \text{Ca}(\text{HCO}_3)_2$, converting insoluble calcium carbonate into soluble calcium bicarbonate. Over millions of years, subterranean caverns, sinkholes, and blind valleys developed. Inside the dark caverns, water droplets saturated with calcium bicarbonate drip from the ceiling. As carbon dioxide degasses from each droplet, insoluble calcium carbonate precipitates. Calcite accumulating downward from the ceiling forms icicle-like 'stalactites', while deposits growing upward from the floor form pillar-like 'stalagmites'. Where the two meet, continuous 'cave columns' are erected.

Q.11 What type of chemical weathering is primarily responsible for carving the Borra Caves?

- (A) Oxidation of iron sulfide ores (B) Carbonation and solution weathering of limestone by carbonic acid
- (C) Thermal exfoliation of granite boulders (D) Hydration of clay silicate minerals

Q.12 How do stalactites form inside subterranean limestone caves?

- (A) Wind blows desert sand into icicle shapes on the roof (B) Calcite precipitates downward from the ceiling as carbon dioxide degasses from dripping groundwater droplets
- (C) Bats construct hanging nests out of river mud (D) Lava cools rapidly upon touching cold cavern air

Q.13 What is formed when a descending stalactite and an ascending stalagmite join together inside a karst cave?

- (A) A cirque (B) A cave pillar or column
- (C) A yardang (D) An alluvial fan

Q.14 A surface depression or funnel-shaped hole formed in karst topography by the collapse of a cave roof is called:

- (A) A sinkhole (or doline) (B) An oxbow lake
- (C) A hanging valley (D) A barchan dune

Q.15 Why does karst topography develop predominantly in limestone and dolomite terrains rather than in quartzite or sandstone?

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|--|--|
| (A) Quartzite is completely soluble in ordinary tap water | (B) Limestone consists of carbonate minerals that react readily with weak carbonic acid, whereas quartz resists acid dissolution |
| (C) Sandstone is much older than any known limestone formation | (D) Limestone only exists in hyper-arid hot desert climates |

CASE STUDY 4: GLACIAL GEOMORPHOLOGY OF THE GANGOTRI GLACIER SYSTEM

The Gangotri Glacier in the Uttarkashi district of Uttarakhand is one of the largest valley glaciers in the Himalayas, extending over 30 kilometres in length and covering an area of approximately 143 square kilometres. Fed by massive snowfields at Chaukhamba peak, the glacier flows down a high-altitude mountain valley. Unlike fluvial streams that cut narrow V-shaped gorges, the immense mass and rigidity of moving glacial ice erodes simultaneously across its entire valley profile via two primary mechanisms: frost plucking (quarrying shattered bedrock) and basal abrasion (striating and polishing bedrock using embedded rock fragments). This transforms pre-existing river valleys into broad U-shaped troughs with steep vertical rock walls and wide, flat floors. Tributary glaciers with less ice volume cannot erode as deeply as the trunk glacier, leaving behind elevated 'hanging valleys' from which waterfalls cascade. At the terminus (Gaumukh), the glacier deposits unsorted glacial till comprising giant boulders, gravel, and clay, constructing massive ridges known as terminal moraines.

Q.16 What cross-sectional shape is characteristic of a valley eroded by an alpine glacier like Gangotri?

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|---------------------------------------|--|
| (A) A deep, razor-thin V-shaped gorge | (B) A wide U-shaped valley with steep vertical walls and a broad floor |
| (C) A circular limestone sinkhole | (D) A flat deltaic coastal plain |

Q.17 What geomorphic landform is created where a smaller tributary glacier joins a deeply carved main trunk glacial valley?

- | | |
|----------------------|---------------------|
| (A) A hanging valley | (B) An oxbow lake |
| (C) A sand spit | (D) A mushroom rock |

Q.18 What are 'moraines' in glacial geomorphology?

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|--|--|
| (A) Sand dunes formed by desert wind gusts | (B) Ridges of unsorted glacial till and debris deposited along the margins and terminus of a glacier |
| (C) Underground rivers flowing through limestone caverns | (D) Coral reefs built in warm tropical lagoons |

Q.19 The snout or terminus of the Gangotri Glacier, from which the Bhagirathi River emerges, is locally called:

- | | |
|-------------|---------------|
| (A) Gaumukh | (B) Bhimbetka |
| (C) Borra | (D) Lothal |

Q.20 What physical process allows moving glacial ice to quarry and lift large blocks of bedrock from the valley floor?

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|---|--|
| (A) Chemical oxidation of quartz | (B) Frost plucking (quarrying), where meltwater freezes into rock joints and tears blocks away as ice advances |
| (C) Thermal exfoliation under blazing desert sunshine | (D) Biological root wedging by desert cactus plants |

CASE STUDY 5: COASTAL MARINE LANDFORMS ALONG THE KONKAN COAST OF MAHARASHTRA

The Konkan coastline of Maharashtra presents an outstanding natural laboratory for studying coastal marine geomorphology shaped by high-energy Arabian Sea waves. Where basaltic rocky headlands jut into the sea, wave refraction concentrates wave energy directly onto the protruding headland promontories. Breaking waves exert intense hydraulic pressure—trapping and compressing air within rock crevices, which explodes outward as the wave retreats. Combined with abrasive rock debris thrown against the cliff base (corrasion), this pounding carves a notch at sea level. Over time, the notch deepens into a sea cave. Where waves attack both sides of a narrow headland, caves meet to form a hollow passageway known as a 'sea arch'. When the roof of the arch eventually collapses under its own weight, an isolated vertical pillar of rock is left standing out at sea, termed a 'sea stack'. As the coastal cliffs continuously retreat landward, a gently sloping bedrock platform is exposed between high and low tide marks, known as a 'wave-cut platform'.

Q.21 What is the correct sequential evolution of coastal erosional landforms on a rocky headland?

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|---|--|
| (A) Sea stack → Sea arch → Sea cave → Delta | (B) Sea cave → Sea arch → Sea stack → Stump |
| (C) Beach → Spit → Lagoon → Cirque | (D) Moraine → U-valley → V-valley → Oxbow lake |

Q.22 What hydraulic mechanism causes explosive mechanical fracturing of coastal cliff crevices by breaking waves?

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|---|---|
| (A) Trapped air inside rock crevices is violently compressed by incoming waves and expands explosively when water recedes | (B) Waves heat the rock until it melts into liquid basalt |
| (C) Marine fish dig deep tunnels through the solid basalt cliffs | (D) Salt crystals freeze into solid ice during tropical summer afternoons |

Q.23 What is an isolated, vertical column of rock left standing offshore after the collapse of a sea arch called?

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|---------------------|------------------|
| (A) A sea stack | (B) A yardang |
| (C) An alluvial fan | (D) A stalagmite |

Q.24 A gently sloping planar bedrock bench carved at the base of a retreating coastal cliff is known as:

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|-------------------------|-------------------------|
| (A) A wave-cut platform | (B) A barchan sand dune |
| (C) A hanging valley | (D) An oxbow lake |

Q.25 Why is wave erosion concentrated predominantly on rocky headlands rather than inside sheltered bays?

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|---|--|
| (A) Wave refraction bends wave crests to converge energy onto headlands while dispersing energy in bays | (B) Bays are located at higher elevations above sea level than headlands |
| (C) Headlands are made of soft butter while bays are made of diamond | (D) Wind never blows inside coastal ocean bays |

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET E)

01	☐ A ☐ B ☐ C ☐ D	02	☐ A ☐ B ☐ C ☐ D	03	☐ A ☐ B ☐ C ☐ D	04	☐ A ☐ B ☐ C ☐ D	05	☐ A ☐ B ☐ C ☐ D
06	☐ A ☐ B ☐ C ☐ D	07	☐ A ☐ B ☐ C ☐ D	08	☐ A ☐ B ☐ C ☐ D	09	☐ A ☐ B ☐ C ☐ D	10	☐ A ☐ B ☐ C ☐ D
11	☐ A ☐ B ☐ C ☐ D	12	☐ A ☐ B ☐ C ☐ D	13	☐ A ☐ B ☐ C ☐ D	14	☐ A ☐ B ☐ C ☐ D	15	☐ A ☐ B ☐ C ☐ D
16	☐ A ☐ B ☐ C ☐ D	17	☐ A ☐ B ☐ C ☐ D	18	☐ A ☐ B ☐ C ☐ D	19	☐ A ☐ B ☐ C ☐ D	20	☐ A ☐ B ☐ C ☐ D
21	☐ A ☐ B ☐ C ☐ D	22	☐ A ☐ B ☐ C ☐ D	23	☐ A ☐ B ☐ C ☐ D	24	☐ A ☐ B ☐ C ☐ D	25	☐ A ☐ B ☐ C ☐ D

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & CASE STUDY ANALYTICAL EXPLANATIONS (SET E)

Q. No.	Correct	Curricular Source Analysis & Conceptual Explanation
Q.1	(B)	The focus is the subterranean point of rupture, while the epicentre is its perpendicular vertical projection on the Earth's surface.
Q.2	(B)	The northward collision of the Indian tectonic plate with Eurasia generates intense compressional stress along faults throughout the subcontinent.
Q.3	(B)	Liquefaction occurs when saturated loose sand loses shear resistance due to elevated pore-water pressure during cyclic seismic shaking.
Q.4	(B)	Primary (P) waves are compressional longitudinal waves that travel fastest and penetrate all phases of matter (solids, liquids, gases).
Q.5	(A)	Unconsolidated, water-saturated soils trap and amplify seismic waves, prolonging ground shaking and increasing structural damage.
Q.6	(B)	In high-gradient upper mountain courses, vertical corrasion dominates, carving steep V-shaped valleys and dramatic canyons.
Q.7	(B)	Braiding occurs when a river's sediment load vastly exceeds its transport capacity upon hitting a low slope gradient, choking the main channel with sandbars.
Q.8	(B)	Majuli is an alluvial riverine island formed by the bifurcating and braiding fluvial dynamics of the Brahmaputra and Subansiri rivers.
Q.9	(B)	High-velocity flood discharge exerts shear stress on outer concave banks, triggering slumping and rapid lateral erosion of unconsolidated riverbanks.
Q.10	(B)	When a meandering river cuts straight through a narrow neck during high discharge, the abandoned curved loop becomes an oxbow lake.
Q.11	(B)	Limestone undergoes carbonation when acidic groundwater dissolves calcium carbonate into soluble calcium bicarbonate.
Q.12	(B)	Stalactites grow downward from cave ceilings through the slow precipitation of calcium carbonate from dripping, degassing mineral water.
Q.13	(B)	When a stalactite growing down meets a stalagmite growing up, they fuse to form a continuous rock column or cave pillar.
Q.14	(A)	Sinkholes (dolines) are circular depressions formed on karst surfaces when acidic dissolution collapses subterranean cavern ceilings.
Q.15	(B)	Carbonate rocks are soluble in naturally acidic water, whereas silicate rocks like quartzite resist carbonic acid dissolution.
Q.16	(B)	Glaciers erode both valley walls and valley floors simultaneously through plucking and abrasion, carving wide U-shaped troughs.
Q.17	(A)	Tributary glaciers erode less deeply than the main trunk glacier; upon melting, their valley floors remain suspended high above as hanging valleys.
Q.18	(B)	Moraines are ridges composed of unsorted, unstratified rock debris (till) transported and dumped by moving or retreating glacial ice.
Q.19	(A)	Gaumukh (cow's mouth) is the terminus of the Gangotri glacier and the traditional source of the Bhagirathi river.

Q. No.	Correct	Curricular Source Analysis & Conceptual Explanation
Q.20	(B)	Glacial plucking occurs when subglacial water freezes into rock fractures, welding blocks to the moving glacier, which rips them from the bedrock.
Q.21	(B)	Marine erosion first carves a sea cave; dual erosion pierces the headland into an arch; roof collapse leaves an isolated stack; further erosion reduces it to a stump.
Q.22	(A)	Hydraulic action traps air within crevices; wave impact compresses it to extreme pressure, and explosive decompression shatters the rock.
Q.23	(A)	A sea stack is an isolated coastal rock pillar left behind after the roof of a sea arch collapses into the sea.
Q.24	(A)	Wave-cut platforms are smooth or pitted rocky benches exposed at low tide, formed by the progressive landward retreat of marine sea cliffs.
Q.25	(A)	Wave refraction bends incoming waves as they feel bottom friction near headlands, focusing concentrated kinetic energy onto headland tips.